CST539 - Software Testing

CST 539 (Software Testing) Syllabus

Course Information

Course Details

- **Course Title:** Software Testing
- **Course Number:** CST 539
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course provides an overview of key software testing techniques and methodologies, focusing on practical applications. Students will learn to design test cases, implement automated scripts, and perform regression and integration testing. Emphasis is placed on adapting testing strategies to agile and DevOps environments using industry-standard tools. Through hands-on exercises, students will gain foundational skills to ensure software quality and reliability.

Materials

- Course textbook: "The Art of Software Testing" by Glenford J. Myers, Corey Sandler, and Tom Badgett

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Design comprehensive test cases and implement automated testing scripts
2. Execute regression and integration testing strategies effectively
3. Adapt testing methodologies to agile and DevOps development environments
4. Utilize industry-standard testing tools to ensure software quality and reliability

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%

Assignment Kind	Value
Participation	30%

Projects

Projects will take the form of test case design exercises, automated testing script development, and focused testing methodology assignments. Students will apply testing techniques to small-scale software projects and develop practical skills in quality assurance.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of code review sessions, testing workshops, peer evaluations of test strategies, and collaborative debugging exercises.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.